Rider pricing and marketplace prediction

Victor H. Aguiar | 26 October 2026 | Narrative memorandum

1. Marketplace context and the decision

Uber operates a physical marketplace in which available capacity moves, trips occupy drivers for different lengths of time, and a decision in one neighborhood affects the next set of opportunities elsewhere. A rider sees a price and expected service, makes a choice, and may still fail to complete a ride because matching, driver acceptance, waiting or cancellation intervenes. The pricing problem therefore concerns both which requests enter the system and how reliably the system can serve them over time.

Several controls interact but should not be conflated. Rider price influences demand; driver compensation influences the attractiveness of serving trips; matching allocates available capacity; product design offers different service-price combinations. Uber publicly distinguishes rider prices from driver earnings and says surge uses market conditions and forecasts. Its matching material describes collective allocation rather than simply assigning every rider the nearest driver. A rider-side experiment must record the other policies and responses rather than assume equivalent driver-pay or supply increases. [1-3] Where offered, Wait & Save exchanges a lower price for a longer wait, while Uber One Surge Savings discounts busy-period rides. Posted fares, member discounts and subsequent credits must remain distinct in response modeling and cost accounting. [4,5]

Conversion needs a denominator. Request acceptance after a quote is one outcome; completion conditional on a request is another. Their product is completed rides per quote opportunity, provided opportunities and observation windows are defined consistently. A higher price can filter out marginal requests, reduce queueing pressure and improve service for those who remain while reducing total completed conversion. Conversely, protecting capacity before a predictable demand increase could avoid later service failures. Which effect dominates depends on behavior, matching and the time horizon. A change that moves a ride to the next interval, another Uber product or a neighboring area is not automatically a net incremental completion.

My economic framing is a constrained dynamic allocation problem. The platform needs a forecast of the relevant market state, a model of how riders and the network respond to candidate actions, and a decision rule with explicit contribution, reliability and longer-run participation objectives. Price is a control in that problem. ETA and completion are consequences of the state and actions, not guaranteed benefits of a higher surge. Aggregate price-volume correlations cannot resolve the trade-off because the same scarcity that causes high prices can also cause long waits and failed requests.

The domain is already technically mature. Uber's published DeepETA work combines routing estimates with learned residuals, while its 2025 matching account uses learned value estimates to consider future network balance. These are examples of existing hybrid prediction and decision systems, not claims about the exact current implementation of every market. They establish the appropriate standard for this proposal: an incremental signal must help a strong incumbent under the same information and serving constraints. A small public benchmark can justify that test; it cannot substitute for it. [6,7]

2. Where a foundation model could contribute

The proposed contribution is transferable prediction, with a narrow interface to the existing decision system. A foundation model brings patterns learned outside one local training sample. That may be useful when a geography or product has little history, when related markets contain information, or when maintaining many separate models is expensive. These are hypotheses about where transfer could pay, not claims that the model is inherently robust to shocks. Our own tail results are a reason to test that proposition carefully rather than to advertise broad generalization.

The first internal target should be chosen with the team that consumes it. Candidate targets include near-term request intensity, serviceable supply, or a residual error in an existing market-pressure forecast. The distinction matters: potential rider arrivals are not observed from completed rides, and serviceable capacity is not a count of every online driver. Driver commitments, travel time and matching eligibility affect what can actually serve a request. A learned residual or additional feature may be more useful than an independent forecast, because Uber's incumbent model already contains valuable local information. The comparison should ask what the foundation model adds after that information is available.

For rider-pricing decisions, a state forecast then needs an action-response model. The useful quantity is the distribution of pickup delay, requesting and completion under a candidate price and specified surrounding policies. Appending a proposed price as a future covariate does not identify this response when historical prices reacted to unobserved scarcity. Logged experiments or another defensible identification strategy are needed to distinguish behavior under different actions. Price-wait preferences can be heterogeneous without implying personalized base fares; common local prices and service menus can allow self-selection. Uber's current U.S. transparency report distinguishes upfront pricing from personalized promotions, which should remain separate in both modeling and accounting. [8]

Economics can improve the learning problem by making the simulated curriculum explicit. A useful curriculum would vary demand shocks, outside options, price and waiting-time sensitivities, queue abandonment, matching efficiency, travel delays and policy responses. It would include situations where higher prices lose riders, where capacity cannot respond quickly, and where a locally attractive action worsens another area or later period. It should separate exogenous surprises from anticipated events and preserve the information available at each decision. The goal is not to encode a preferred pricing conclusion; it is to expose the predictor to meaningful state transitions and known failure modes.

Serving must respect the distinction between forecast cadence and quote latency. A candidate design refreshes a forecast or residual asynchronously, publishes it with freshness and uncertainty metadata, and lets the existing low-latency service consume a cached value or fall back to its incumbent input. That is an engineering proposal to validate, not a description of Uber's private stack. An hourly public-data pilot says little about millisecond quote serving or a five-minute control loop. The value of my experience with tabular and time-series foundation models, structural consumer behavior and industrial forecasting is in connecting these choices and owning the test with engineering, rather than treating a better model score as a completed system.

3. What the public data experiment tested

I began with a question that public data could answer: can an unadapted time-series foundation model forecast a structured rider-fare series competitively against credible local baselines? I reused my GARP-EFM infrastructure on a local Apple M1 and processed 62,874,417 January-March 2026 NYC TLC records. Filtering and January-only route selection retained 237,041 eligible completed Uber trips on six short Brooklyn and Queens origin-destination pairs. The declared Manhattan density screen did not produce six qualifying routes, so the documented NYC fallback applied. This is a small, selected market slice, not a representative sample of Uber's global business. [9]

The outcome is hourly median base_passenger_fare before tips, tolls, taxes and fees. The field name does not mean a no-surge counterfactual. There is no explicit surge component, product-tier identifier, rejected-quote population, rider membership status or available-driver state in this panel. Consequently, a successful forecast predicts an outcome of the historical pricing policy and trip mix. It does not estimate elasticity, reconstruct the pricing algorithm or reveal the price that would maximize completion or contribution. Fixing the target this way keeps the empirical question answerable while preserving the distinction between the experiment and the business problem.

The evaluation uses January and early February for training, the second half of February for chronological selection, and March 1-28 for testing. Routes are selected before validation and test processing. The original Chronos-2 weights are hash-identical to their October 2025 release, before these outcomes. This chronology matters because older taxi series occur in the model's pretraining inventory. Every method uses the same observed target hours and a completed-trip availability delay: the newest usable pickup-hour bin has ended at least three hours before the forecast origin. Low-volume target hours remain missing rather than becoming fabricated zeros. [10]

Six baselines cover persistence, daily and weekly repetition, a four-week seasonal estimate, regularized lagged regression, and tuned histogram gradient boosting. The learned models receive causal fare/count lags, rolling features, route identifiers and known calendar information. Hyperparameters are selected on validation and the selected learned models are refitted on available training-plus-validation labels. Original Chronos variants use either fare history alone or fare history with calendar and past completed-trip counts. The context is 672 hours, with first-, third- and sixth-hour target bins and six-hour origin spacing. This is a controlled information comparison, not a claim to exhaust all classical methods.

On 1,574 matched test cases, the covariate-informed zero-shot model achieved $1.066787 mean absolute error, versus $1.177008 for tuned boosting, a 9.36% reduction. The validation-selected non-foundation reference was four-week seasonality, whose test error was $1.310977; the corresponding reduction was 18.63%. The univariate zero-shot model achieved $1.090801. The average advantage over boosting held across the six routes and three reported horizons. These results demonstrate useful zero-shot predictive performance on this task. They do not compare with Uber's internal models. Different horizons cover different target-clock samples, so their relative errors are not an experiment on how difficulty changes with forecast distance.

4. What we learned about decisions and adaptation

The average improvement is not sufficient for surge use. In 143 high-fare cases, defined by training-only route thresholds, the covariate model's error was $2.550145 versus $2.476766 for the seasonal reference, about 2.96% worse. Its nominal 80% intervals covered 76.56% of outcomes. The largest miss was a forecast of approximately $11.38 against a $29.13 realized median on Astoria-Steinway; several routes missed sharply around the same time. No cause is assigned without evidence about events or composition. The diagnostic directs the next evaluation toward tail loss, uncertainty and regime changes, where a pricing consumer may care more than it does about average dollar error.

I also built a stylized hourly simulator to make the economic structure explicit. Potential arrivals and capacity vary; riders have heterogeneous positive sensitivities to price and quoted wait; requests may abandon; existing queued requests keep their quoted fares; and service is capacity constrained. Rider price does not mechanically create capacity or change driver compensation. The qualitative motivation comes from Buchholz and coauthors' Econometrica work on price-wait preferences, not from importing its Liftago estimates. The simulator's parameters are illustrative, with only coarse fare and activity scales anchored to the real training period. It is a training and reasoning environment, not an estimated Uber market. [11]

A bounded adaptation test trained a fresh rank-eight LoRA adapter on synthetic fare trajectories for 60 CPU steps, saving checkpoints at 30 and 60. Base weights stayed fixed, and merged models reproduced adapter predictions. The diagnostic 30-step checkpoint reached $1.066502 test MAE, only 0.027% below the unadapted covariate model; real validation preferred no adaptation. This is not a meaningful demonstrated curriculum gain. It means the local training and evaluation path works, while a better curriculum still has to prove itself. Repeating longer training or changing the curriculum after seeing this test would require a new declared experiment rather than reusing the same result as confirmation.

To separate forecasting from acting, a second synthetic experiment jointly predicts next-hour potential arrivals and service capacity. It uses 64 fixed cases from 16 held-out trajectories. Domain-trained regression and

boosting receive separate synthetic training examples; Chronos receives no domain gradient updates. A declared heterogeneous-demand and congestion calculator then evaluates candidate price multipliers using the forecast state and current queue. Its reference response is the same calculator evaluated with realized synthetic inputs. These are model-implied responses, not observed conversion or ETA from the original simulation and not empirical pricing effects.

Zero-shot joint-state MAE was 5.3960 versus 5.3771 for boosting, essentially competitive with a small disadvantage. Modeled ETA-response error was 1.2896 versus 1.2988 minutes, and modeled completion-response error was 5.6896 versus 5.9065 percentage points. But the point-forecast controller is not safe: Chronos abstained in 17 of 64 cases, and 13 of its 47 chosen actions violated the ETA constraint under realized-input evaluation. Boosting had comparable failures. This is the substantive lesson of the prototype: reasonably accurate state predictions do not by themselves produce reliable constrained decisions. Uncertainty, misspecification and action-response validation are first-order work, not final deployment polish.

5. Business value and the engineering contract

The business opportunity is the value of better decisions relative to the current production policy, after costs and displacement. It is not the dollar size of a fare forecast error, and a 9.4% reduction in that error is not a 9.4% gain in revenue or conversion. A forecast may help anticipate a capacity imbalance, select a less costly intervention, improve the timing of an existing action, or avoid an unreliable price recommendation. It can also have no incremental value if the incumbent already captures its signal or if another constraint binds. The economic test must therefore be attached to a specific consuming decision and measured outcome.

For scale only, Uber reported $28.988 billion of Mobility Gross Bookings in Q2 2026. Four times that quarter is $115.952 billion, not an annual forecast. The illustration assumes that 20% is eligible, baseline quote-to-completion conversion is 60%, and incremental completed volume contributes 10% of baseline gross fare after relevant variable costs. It assumes $1 million annual all-in program cost, unchanged average gross fares and no extra subsidy. These are scenario inputs, not Uber operating estimates. Let G be eligible gross bookings, c baseline conversion and d the net absolute conversion lift. Under those assumptions, incremental contribution before program cost is G times d/c times the incremental-volume margin. [12]

A lift of 0.05 percentage points means five additional completed rides per 10,000 eligible quote opportunities, after cannibalization and time-shifting. The scenario produces approximately $19.33 million additional gross bookings, $1.93 million contribution and $0.93 million after program cost. At zero lift, the program loses its $1 million cost. Break-even is approximately 2.59 extra completions per 10,000 quotes; a $2 million program cost makes the five-completion scenario slightly negative. The interpretation is a hurdle for an internal experiment, not a valuation established by the demo. These effects must be incremental to Uber's incumbent system, not to our public-data baselines.

Price and subsidy changes require separate accounting. Cutting the average gross fare affects existing eligible rides as well as incremental volume; the additional conversion can fail to cover that loss. Consumer coupons are costs and do not mechanically reduce Uber's reported pre-discount Gross Bookings. Driver compensation, subscription credits and promotions also change the contribution calculation. The included calculator exposes these channels rather than hiding them inside the forecast gain. A serving-cost extrapolation from batched laptop inference is only a capacity illustration: it is not a cloud quote, a reliability budget or evidence that the science and engineering work will cost almost nothing.

The science-engineering contract should start with the consumer of the forecast. We need agreed target semantics, geographical units, update cadence, horizon, features, availability timestamps and a fallback. Science owns the model and economic evaluation with engineering, including data invariants and failure analysis; engineering collaboration covers interfaces, state freshness, caching, load, observability and rollout behavior. Outputs should include an age and uncertainty indicator, not just a number. When an upstream feed

is late or the state leaves supported conditions, the service should use a declared fallback rather than let a plausible-looking stale forecast silently steer price. Ownership continues through validation, rollout and monitoring.

6. Recommendation and the internal experiment

I recommend a bounded internal evaluation of one decision-relevant state or residual forecast, selected with the rider-pricing and engineering teams. The initial deliverable is a replay and shadow comparison against the actual incumbent with identical feature availability, geography, cadence and horizons. Quote sessions that do not become requests, completed and failed requests, quoted and realized pickup times, driver offers and availability, and the surrounding pricing/matching policies are needed to define the population. We should not translate the hourly public task directly into a real-time pricing target without that agreement.

The first gate is incremental predictive usefulness, including tails, calibration, shifts, freshness and serving performance. If there is no useful signal beyond the current system, stop or retain the model only as a diagnostic benchmark. If the signal helps, test its effect on an existing decision mechanism before expanding the model's authority. Economic uncertainty and forecast uncertainty should be propagated into constrained decisions; the toy controller's failures argue against acting on point estimates alone. A marketplace curriculum becomes a separate experiment with held-out regimes, explicit information sets and an unadapted-model control, not an assumed source of additional lift.

A subsequent causal test needs units and horizons that address network interference. Rider-level randomization alone may move available supply between treated and control opportunities; temporal alternation can retain queue and repositioning effects. Geography, timing, carryover and washout must be designed for the intervention rather than selected by habit. The primary business metric should reflect net contribution and completed rides per consistently defined opportunity, with rider reliability, driver outcomes and longer-run participation as guardrails. This is the point at which the scenario's break-even target becomes an empirical question.

The proposition for Hossein is an end-to-end contribution: bring experience with structured-data foundation models and economic behavior, define a consuming decision with engineering, establish whether a new predictive signal survives a credible comparison, and carry a useful signal through a safe operational test. The work so far supplies a reproducible positive forecasting result, an adaptation test with no meaningful gain and a concrete example of why prediction quality is not policy safety. Together they justify a focused technical discussion about the next internal test, rather than a claim that a laptop demo has solved surge pricing.

Public sources and supporting evidence

[1] Uber Marketplace Pricing. Rider prices and driver earnings are distinct.

[2] Uber Marketplace Surge Pricing. Local conditions and forecasts already inform surge.

[3] Uber Marketplace Matching. Matching is a collective allocation problem.

[4] Uber Wait & Save. Lower price in exchange for a longer wait, where offered.

[5] Uber One U.S. benefits, October 2025. Surge Savings discounts during busy periods.

[6] DeepETA, February 2022. Published hybrid ETA architecture; not asserted to be the unchanged current system.

[7] Uber Marketplace Balance, July 2025. Future network balance in published matching work.

[8] Uber U.S. Algorithmic Transparency Report 2026. Pricing and promotion distinctions, printed pp. 8-11.

[9] NYC TLC trip records and HVFHV dictionary. Official source; exact filters and exclusions in the repository.

[10] Chronos-2 technical report. Original model and training inventory; pinned weight provenance in the repository.

[11] Buchholz et al., Econometrica 2025. Price-wait preference motivation; no transfer of estimated parameters.

[12] Uber Q2 2026 results. Mobility Gross Bookings scale only.

Experiment protocols, predictions and independent audits: https://github.com/vaguiarl/marketplace-foundation-forecasting.